Notes: Hendershott, Jones, and Menkveld (JF, 2011)

Does Algorithmic Trading Improve Liquidity?

Contents

1 Big picture

- **Question.** Does algorithmic trading (AT) improve market liquidity, or does it instead intensify adverse selection and make markets harder for conventional liquidity suppliers?
- **Setting.** The paper studies NYSE common stocks in the post-decimalization period and focuses on a 2003 market-design change called *autoquote*, which automated dissemination of the inside quote.
- **Core empirical challenge.** AT is endogenous. Traders choose whether to adopt algorithms, and they may do so precisely when liquidity, volatility, or order-flow conditions change. A simple correlation between AT and spreads therefore cannot be given a causal interpretation.
- **Main empirical strategy.** The paper instruments for AT using the staggered introduction of autoquote across NYSE stocks. Autoquote generated a plausibly exogenous increase in algorithm-friendly quote updates, especially for actively traded large-cap stocks.
- **Main result.** For large stocks, more AT narrows quoted and effective spreads, lowers adverse selection, reduces the amount of price discovery that arrives through trades, and increases the informativeness of quotes.
- **Important nuance.** Liquidity supplier revenues, measured by realized spreads, rise temporarily after the introduction of autoquote. So the improvement in liquidity does *not* come from lower liquidity-provider revenues. Instead, it comes from a larger fall in adverse-selection losses.
- **Where the evidence is strongest.** The cleanest and most reliable effects are in large- and medium-cap stocks. For the smallest stocks, the instrument is weaker and the estimates are much less precise.
- **Why the paper matters.** This is one of the first causal empirical papers on AT. It moves the literature beyond simple time-series coincidence and asks whether a trading-technology innovation truly changes market quality.

2 Data and measurement (key objects)

2.1 Long-run time-series sample

- **Purpose.** Describe the joint evolution of AT and liquidity over time.
- **Sample period.** February 2001 through December 2005, entirely in the post-decimalization regime.
- **Stock universe.** NYSE common stocks matched in TAQ and CRSP. The authors impose a balanced-panel requirement so that the same stocks are followed throughout the sample.
- **Price filters.** Stocks with average prices below \$5 or above \$1,000 are excluded.
- **Final sample.** 943 stocks observed monthly.
- **Sorting.** Stocks are sorted into market-cap quintiles, where quintile 1 contains the largest-cap stocks and quintile 5 the smallest-cap stocks.

2.2 Event-study / IV sample around autoquote

- **Purpose.** Estimate the causal effect of AT using the staggered adoption of autoquote.
- **Sample period.** December 2, 2002 through July 31, 2003, which gives roughly two months before and two months after the 2003 rollout window.
- **Balanced daily panel.** The paper keeps stocks with at least 21 trades per day for each day in the 8-month sample.
- **Final sample.** 1,082 NYSE common stocks.
- **Why a separate sample is needed.** The instrumental-variables exercise requires daily variation around the exact rollout dates, whereas the earlier figures are intended only to describe broader trends.

2.3 Data sources and observed variables

- **TAQ.** Used for trades, quotes, quoted depth, spreads, and trade signing.
- **CRSP.** Used for prices, market capitalization, and volatility-related measures.
- **NYSE System Order Data (SOD).** Used for electronic message traffic and specialist participation.
- **Control variables.** The main controls in the IV specifications are share turnover, volatility, inverse price, and log market capitalization.
- **Additional outcomes.** The paper also studies trade size, trades per minute, and specialist participation rates.

2.4 Proxy for algorithmic trading

The paper cannot directly observe whether a given order was generated by an algorithm. It therefore uses *electronic message traffic* as a proxy for AT.

- **Message definition.** NYSE electronic messages include electronic order submissions, cancellations, and trade reports handled by SuperDOT and recorded in SOD.
- **What is excluded.** The measure excludes specialist quoting and manual orders sent to the floor and handled by floor brokers.
- **Normalized AT proxy.** Let Volume_{it} denote stock i's dollar volume on day t, and let Messages_{it} denote the number of electronic messages. The paper's preferred AT proxy is

$$\text{algo\_trad}_{it} = -\frac{\text{Volume}_{it}/100}{\text{Messages}_{it}}.$$

- **Interpretation.** A higher value of algo_trad_{it} means *less* dollar volume per message and therefore *more* messages per dollar traded. In other words, higher values correspond to more intensive algorithmic activity.

- **Why normalization matters.** Raw message counts rise with trading activity. By normalizing by volume, the paper tries to isolate changes in the *nature* of trading rather than simple changes in the amount of trading.
- **Economic meaning.** Because variation in the normalized measure mainly reflects submissions and cancellations rather than trade reports, it primarily captures algorithmic liquidity supply and order management.

2.5 Liquidity measures

For stock j and transaction t, let $q_{jt} = +1$ for buyer-initiated trades and $q_{jt} = -1$ for seller-initiated trades. Let p_{jt} be the trade price and m_{jt} the prevailing quote midpoint.

- **Quoted half-spread.** Share-weighted average quoted half-spread in basis points.
- **Quoted depth.** Share-weighted quote depth measured in thousands of dollars.
- **Effective half-spread.** The proportional effective half-spread is

$$\text{espread}_{jt} = \frac{q_{jt}(p_{jt} - m_{jt})}{m_{jt}}.$$

- **Realized half-spread.** Using the midpoint 5 minutes later,

$$\text{rspread}_{jt} = \frac{q_{jt}(p_{jt} - m_{j,t+5\text{ min}})}{m_{jt}}.$$

  The paper also computes a 30-minute version.

- **Adverse selection / price impact.** The 5-minute price impact is

$$\text{adv\_selection}_{jt} = \frac{q_{jt}(m_{j,t+5\text{ min}} - m_{jt})}{m_{jt}},$$

  with an analogous 30-minute version.

- **Identity.** By construction,

$$\text{espread}_{jt} = \text{rspread}_{jt} + \text{adv\_selection}_{jt}.$$

- **Trade signing.** The paper follows Lee and Ready (1991) and uses contemporaneous quotes to sign trades.

2.6 Unobservables and empirical applications

- **Main unobservable.** The true algorithmic origin of an order or cancellation is not observed directly.
- **Another unobservable.** The informational content of quotes versus trades is not directly observed; it is inferred using the Hasbrouck VAR decomposition.
- **Empirical applications.** The paper studies:

- whether autoquote actually increases AT,
- whether more AT narrows spreads,
- whether liquidity improves through lower adverse selection or lower liquidity-provider revenues,
- whether AT changes trade size and specialist participation,
- whether AT shifts price discovery away from trades and toward quotes.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline liquidity relation and the endogeneity problem

A natural panel regression would relate liquidity L_{it} to AT A_{it} and controls X_{it}:

$$L_{it} = \alpha_i + \beta A_{it} + \delta' X_{it} + \varepsilon_{it}.$$

- **Interpretation.** This equation asks whether stocks are more liquid when algorithmic activity is higher.
- **Problem.** A_{it} is endogenous. Traders may adopt or intensify AT precisely when liquidity changes, so OLS does not identify the causal effect of AT on liquidity.
- **Paper's solution.** Use a market-design shock—autoquote—as an instrument for A_{it}.

3.2 First-stage regression for the instrument

To verify that autoquote increases AT, the paper estimates

$$M_{it} = \alpha_i + \gamma_t + \beta Q_{it} + \varepsilon_{it},$$

where M_{it} can be messages per minute, the normalized AT proxy, or other daily variables, and Q_{it} is an indicator equal to one after stock i moves to autoquote.

- **Stock fixed effects α_i.** Remove persistent differences across firms, such as average liquidity or average algorithmic activity.
- **Day effects γ_t.** Remove market-wide conditions, such as broad changes in volatility or trading volume.
- **Interpretation.** This is the difference-in-differences style first stage: once stock and day effects are included, identification comes from the staggered timing of autoquote across stocks.

3.3 Main IV liquidity specification

The core specification is

$$L_{it} = \alpha_i + \gamma_t + \beta A_{it} + \delta' X_{it} + \varepsilon_{it},$$

where A_{it} is the endogenous AT measure algo_trad_{it}, instrumented by $Q_{it} = \text{auto\_quote}_{it}$.

- **Dependent variables.** Quoted spread, quoted depth, effective spread, realized spread, and adverse selection.
- **Controls.** Share turnover, Parkinson volatility, inverse price, and log market capitalization.
- **Separate estimation by size quintile.** The paper expects AT to matter more in active large-cap stocks, so it estimates the model separately by quintile.
- **Interpretation of the AT coefficient.** Since A_{it} is the negative of volume per message, a positive shift in A_{it} means more messages per dollar traded. A negative β in a spread regression therefore means that more AT narrows spreads.

3.4 Spread decomposition regressions

Using the identity

$$\text{espread}_{jt} = \text{rspread}_{jt} + \text{adv\_selection}_{jt},$$

the paper runs the same IV specification with realized spread and adverse selection on the left-hand side.

- **Interpretation.** This decomposes the effect of AT on effective spreads into
  - liquidity-provider revenues, and
  - losses associated with adverse selection.
- **Economic question.** Does AT improve liquidity by making liquidity provision more competitive, or by helping liquidity suppliers avoid being picked off?

3.5 Hasbrouck VAR for price discovery

To study whether AT changes the *nature* of price discovery, the paper uses the Hasbrouck (1991a,b) framework. Let r_t be the quote-midpoint return from trade $t-1$ to trade t, and let q_t be the signed trade indicator. The VAR is

$$r_t = \sum_{i=1}^{10} \alpha_i r_{t-i} + \sum_{i=0}^{10} \beta_i q_{t-i} + \varepsilon_t^r,$$

$$q_t = \sum_{i=1}^{10} \gamma_i r_{t-i} + \sum_{i=1}^{10} \phi_i q_{t-i} + \varepsilon_t^q.$$

The VAR is inverted into a vector moving-average representation, and the variance of the random-walk component of prices can be decomposed into

$$\sigma_w^2 = \left(\sum_{i=0}^{\infty} a_i \right)^2 \sigma_r^2 + \left(\sum_{i=0}^{\infty} b_i \right)^2 \sigma_q^2.$$

- **Trade-unrelated component.** The first term captures quote changes that are orthogonal to recent trades.
- **Trade-related component.** The second term captures the share of efficient-price variance associated with recent trades.

- **Permanent price impact of a trade.** The cumulative impulse response $\sum_{i=0}^{\infty} b_i$ measures the long-run price response to an order-flow shock.
- **Interpretation.** If AT updates quotes more quickly, then less information should arrive through trades and more should arrive directly through quotes.

3.6 Nonspread regressions

The paper also runs the IV specification with nonspread outcomes on the left-hand side:

$$M_{it} = \alpha_i + \gamma_t + \beta A_{it} + \varepsilon_{it},$$

where M_{it} is share turnover, trades per minute, trade size, or specialist participation.

- **Trade size.** Helps assess whether algorithms "slice and dice" large orders into smaller pieces.
- **Specialist participation.** Helps identify whether the rents created by autoquote accrue to specialists or to other liquidity providers.

4 Identification and instruments (what makes the estimates credible)

4.1 Why standard correlations are not enough

- AT is a choice variable. Traders adopt algorithms when the benefits are high, and those benefits may depend on liquidity and volatility.
- A simple time-series plot shows AT rising while spreads fall, but that is only suggestive.
- The paper explicitly argues that humans may be used more when markets are illiquid and volatile, so reverse causality is a real concern.

4.2 What autoquote changes

Before autoquote, specialists were responsible for manually disseminating the NYSE inside quote. Autoquote automatically disseminated a new inside quote whenever the limit order book changed in a relevant way.

- **Examples of relevant changes.** A better-priced order arrives, an inside order is canceled, the inside quote is hit or lifted, or inside size changes.
- **What did *not* change.** Specialists could still manually disseminate quotes and still retained their other institutional advantages. NYSE market share stayed roughly unchanged around 80%.
- **Why this matters for algorithms.** Faster quote updates provide immediate feedback about market conditions. A few seconds of speedup are important for algorithms but much less likely to alter the behavior of slower human traders directly.

4.3 Staggered rollout and identification

- **Rollout dates.** Autoquote started on January 29, 2003 with six large, active stocks. More than 200 additional stocks were phased in over the next two months, and all remaining NYSE stocks were migrated on May 27, 2003.
- **Early adopters.** Early stocks tended to be large and active because the NYSE believed these were the securities that would benefit most from the new liquidity-quote framework.
- **Additional selection margin.** Conditional on being early candidates, conversations with NYSE personnel suggest that the specialist's receptiveness to new technology also influenced rollout order.
- **Why staggered timing helps.** Once stock fixed effects and day effects are included, the design compares already-autoquoted stocks to not-yet-autoquoted stocks on the same calendar day.

4.4 Why the exclusion restriction is plausible

- The rollout schedule was fixed months in advance, which makes it hard to believe that migration dates were chosen in response to contemporaneous stock-specific liquidity shocks.
- The exclusion restriction does *not* require the rollout to be random. It only requires that rollout timing not be correlated with *idiosyncratic* liquidity changes that are unrelated to AT and not absorbed by the controls or fixed effects.
- The main threat would be sufficiently persistent temporary shocks. For example, if temporarily illiquid stocks were deliberately chosen early and remained unusually illiquid at the adoption date, post-adoption mean reversion could be misread as an AT effect.
- The paper checks this by estimating AR(1) dynamics for effective spreads and finds an average coefficient around 0.18, implying a half-life of less than one day. This makes long-lasting idiosyncratic liquidity shocks unlikely.
- The paper also reports that predicted liquidity just ahead of migration is not significantly different from unconditional mean liquidity.

4.5 First-stage strength

- Autoquote clearly increases message traffic and the normalized AT proxy, especially for large-cap stocks.
- The first-stage F-statistics reported in the main IV tables range from 5.88 to 7.32, and the null that the instruments do not enter the first stage is strongly rejected.
- The instrument is weaker for some smaller-cap stocks because many of them were migrated together near the end of the rollout, leaving less staggered variation for identification.

4.6 What remains untestable

- The paper cannot directly test that autoquote affects liquidity *only* through AT.

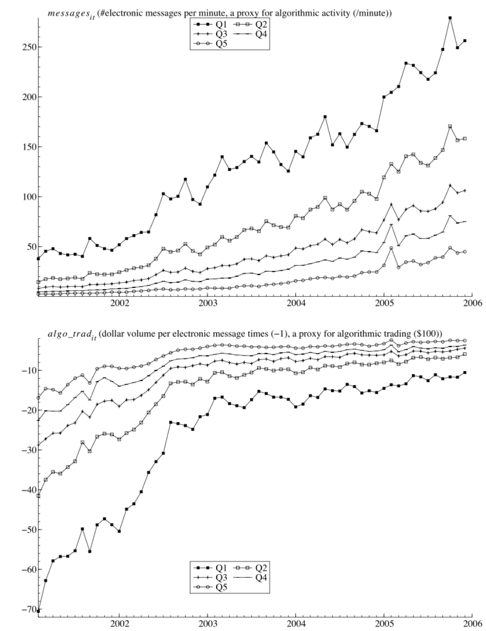

Figure 1. Algorithmic trading measures. For each market-cap quintile, where Q1 is the largest-cap quintile, these graphs depict (i) the number of (electronic) messages per minute and (ii) our proxy for algorithmic trading, which is defined as the negative of trading volume (in hundreds of dollars) divided by the number of messages.

- The authors argue that the few-second speed improvement should matter mainly for algorithms, not for human traders, and that this is the economically relevant channel.
- The data also do not reveal who exactly is trading algorithmically at the order level, so the paper cannot directly identify whether the benefiting liquidity suppliers are proprietary firms, institutions using execution algorithms, or some other participants.

4.7 Relation to the literature

- The paper sits at the intersection of market microstructure, limit-order trading, and technology adoption in financial markets.
- Relative to older spread models, its key contribution is empirical identification: it brings a trading-technology shock to the data and asks what it does to actual liquidity outcomes.
- The paper also links to Hasbrouck-style price discovery work by asking not just whether prices become more informative, but *how* that information gets into prices—through trades or through quote revisions.

5 Tables and figures

5.1 Figure 1 and Figure 2: The basic time-series facts

Read:

- Algorithmic activity rises sharply from 2001 to 2005. For the largest-cap quintile, raw electronic messages increase from roughly 35 per minute just after decimalization to roughly 250 per minute by the end of 2005.

- The normalized measure tells the same story: large-cap stocks move from about $7,000 of trading volume per message to about $1,100 per message, meaning far more messages per dollar traded.
- Over the same period, quoted and effective spreads generally fall.
- These figures motivate the paper but do *not* identify causality, because many other things are changing over the same period.

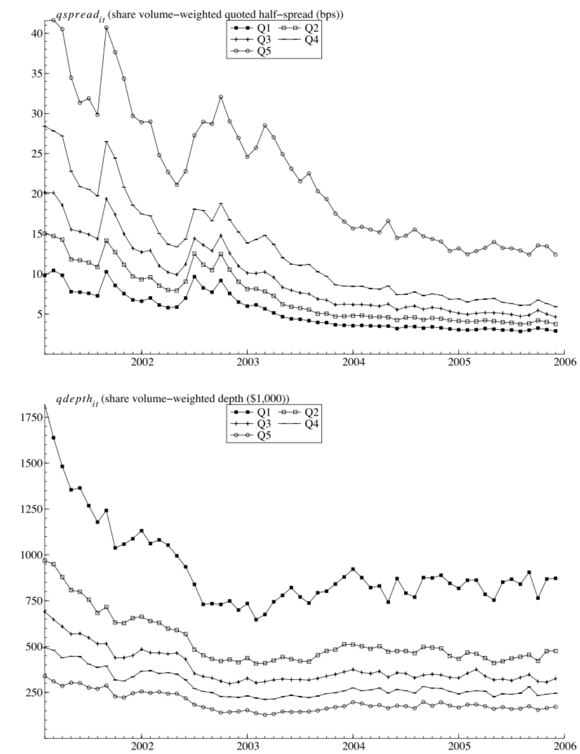

Figure 2. Liquidity measures. These graphs depict (i) quoted half-spread, (ii) quoted depth, and (iii) effective spread. All spread measures are share volume-weighted averages within-firm that are then averaged across firms within each market-cap quintile, where Q1 is the largest-cap quintile.

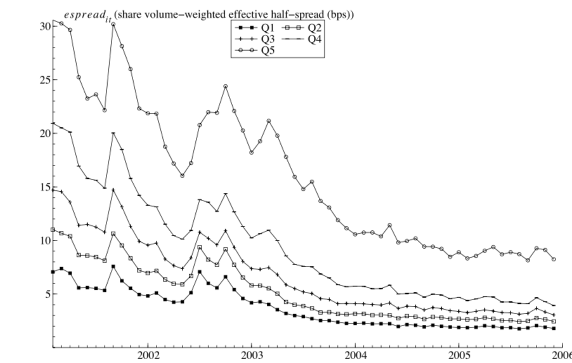

Figure 2. Continued

5.2 Table I: Summary statistics

Read:

- Market quality differs sharply across the size distribution. In the event-study sample, mean quoted half-spreads rise from **5.19 bps** in quintile 1 to **19.89 bps** in quintile 5, while effective half-spreads rise from **3.63 bps** to **14.50 bps**.
- Large-cap stocks also have much greater message intensity: about **119 messages per minute** in quintile 1 versus only about **10** in quintile 5.
- Quoted depth falls steeply with size, from about **$71.2 thousand** in quintile 1 to about **$15.8 thousand** in quintile 5.
- The summary statistics justify the paper's decision to estimate the main specifications separately by market-cap quintile.

Table I
Summary Statistics

This table presents summary statistics on daily data for the period December 2002 through July 2003. This period covers the phase-in of autoquote, used as an instrument in the instrumental variable analysis. The data set combines TAQ, CRSP, and the NYSE System Order Data (SOD) database. The balanced panel consists of 1,082 stocks sorted into quintiles based on market capitalization, where quintile 1 contains largest-cap stocks. All variables are 99.9% winsorized. The within standard deviation is based on day t's deviation relative to the time mean, that is, $x^*_{it} = x_{it} - \bar{x}_i$.

Variable	Description (Units)	Source	Mean Q1	Mean Q2	Mean Q3	Mean Q4	Mean Q5	St. Dev. Within
$qspread_{it}$	share volume-weighted quoted half-spread (bps)	TAQ	5.19	6.82	9.17	11.68	19.89	4.84
$qdepth_{it}$	share volume-weighted depth ($1,000)	TAQ	71.22	41.85	31.43	24.12	15.76	23.42
$espread_{it}$	share volume-weighted effective half-spread (bps)	TAQ	3.63	4.79	6.56	8.46	14.50	3.73
$rspread_{it}$	share volume-weighted realized half-spread, 5min (bps)	TAQ	1.21	1.44	1.88	1.97	4.34	4.71
$adv\_selection_{it}$	share volume-weighted adverse selection component half-spread, 5min, "effective-realized" (bps)	TAQ	2.42	3.35	4.69	6.50	10.16	5.12
$messages_{it}$	#electronic messages per minute, a proxy for algorithmic activity (/minute)	SOD	119.30	53.90	29.81	19.33	10.44	15.55
$algo\_trad_{it}$	dollar volume per electronic message times (−1), a proxy for algorithmic trading ($100)	TAQ/SOD	−18.44	−10.99	−8.05	−6.39	−4.61	4.54
$dollar\_volume_{it}$	average daily volume ($million)	TAQ	94.71	24.09	10.12	5.32	2.17	22.72
$trades_{it}$	#trades per minute (/minute)	TAQ	5.72	2.92	1.78	1.24	0.72	0.72
$share\_turnover_{it}$	(annualized) share turnover	TAQ/CRSP	1.11	1.52	1.48	1.45	1.30	1.16
$volatility_{it}$	standard deviation open-to-close returns based on daily price range, that is, high minus low, Parkinson (1980), (%)	CRSP	1.47	1.56	1.63	1.74	2.06	0.85
$price_{it}$	daily closing price ($)	CRSP	40.01	32.05	25.86	23.93	16.41	3.46
$market\_cap_{it}$	shares outstanding times price ($billion)	CRSP	28.99	4.09	1.71	0.90	0.41	1.96
$trade\_size_{it}$	trade size ($1,000)	TAQ	37.56	19.41	13.06	9.73	6.61	8.03
$specialist\_particip_{it}$	specialist participation rate (%)	SOD	13.07	12.97	13.08	13.73	15.84	3.92

observations: 1,082 ∗ 167 (stock ∗ day)

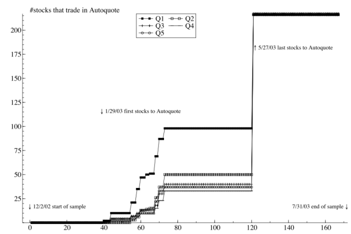

Figure 4. Autoquote introduction. This graph depicts the staggered introduction of autoquote on the NYSE. It graphs the number of stocks in each market-cap quintile that are autoquoted at a given time. Quintile 1 contains largest-cap stocks.

Table II
Autoquote Impact on Messages, Algorithmic Trading Proxy, and Covariates

This table shows the impact of autoquote on other variables, and the second column can be interpreted as the first-stage instrumental variables regression when $algo\_trad_{it}$ is the dependent variable. The analysis is based on daily observations from December 2002 through July 2003, which covers the phase-in of autoquote. We regress each of the variables used in the IV analysis on the autoquote dummy ($auto\_quote_{it}$) using the following specification:

$$M_{it} = \alpha_i + \gamma_t + \beta Q_{it} + \varepsilon_{it},$$

where M_{it} is the relevant dependent variable, for example, the number of electronic messages per minute, Q_{it} is the autoquote dummy set to zero before the autoquote introduction and one afterward, α_i is a stock fixed effect, and γ_t is a day dummy. There are also separate regressions for each size quintile, and statistical significance is based on standard errors that are robust to general cross-section and time-series heteroskedasticity and within-group autocorrelation (see Arellano and Bond (1991)). Table I provides other variable definitions. */** denote significance at the 95%/99% level.

	$messages_{it}$	$algo\_trad_{it}$	$share\_turnover_{it}$	$volatility_{it}$	$1/price_{it}$	$ln\_market\_cap_{it}$
Slope coefficient from regression of column variable on $auto\_quote_{it}$						
All	2.135**	0.291**	−0.016**	0.001	0.000**	−0.003**
Q1 (largest cap)	6.286**	0.414**	0.016*	−0.003	0.000**	−0.005**
Q2	0.880**	0.396**	−0.029*	0.007	−0.000**	0.003**
Q3	0.944**	0.292**	0.002	−0.001	0.000	−0.004**
Q4	0.223**	0.029	−0.006	−0.003	−0.000	0.002
Q5 (smallest cap)	−0.031	0.219**	−0.080**	0.003	0.002**	−0.013**

5.3 Figure 4 and Table II: Autoquote is a valid first-stage shock

Read:

- Figure 4 shows the staggered rollout. Large, active stocks move first; smaller stocks are concentrated later in the rollout.
- Table II shows that autoquote raises message traffic and raises the normalized AT proxy. In the full sample, autoquote increases messages by about **2.135 per minute**; in the largest-cap quintile, the increase is about **6.286 per minute**.
- The first-stage effect on algo_trad_{it} is also positive and significant overall (**0.291**) and especially strong for large caps (**0.414** in quintile 1).
- By contrast, there is no consistent pattern in the other covariates. That is exactly what the paper wants: autoquote shifts AT, not every market characteristic at once.

5.4 Table III Panel A: Baseline liquidity results

Table III
Effect of Algorithmic Trading on Spread

This table regresses various measures of the (half) spread on our algorithmic trading proxy. It is based on daily observations from December 2002 through July 2003, which covers the phase-in of autoquote. The nonsynchronous autoquote introduction instruments for the endogenous $algo\_trad_{it}$ to identify causality from algorithmic trading to liquidity. The specification is

$$L_{it} = \alpha_i + \gamma_t + \beta A_{it} + \delta X_{it} + \varepsilon_{it},$$

where L_{it} is a spread measure for stock i on day t, A_{it} is the algorithmic trading measure $algo\_trad_{it}$, and X_{it} is a vector of control variables, including share turnover, volatility, 1/price, and log market cap. Coefficients for the control variables and time dummies are quintile-specific. Market cap–weighted coefficients are reported for the control variables. Fixed effects and time dummies are included. The set of instruments consists of all explanatory variables, except that $algo\_trad_{it}$ is replaced with $auto\_quote_{it}$. There are separate regressions for each size quintile, and t-values in parentheses are based on standard errors that are robust to general cross-section and time-series heteroskedasticity and within-group autocorrelation (see Arellano and Bond (1991)). */** denote significance at the 95%/99% level. In Panel B the suffix indicates the effective spread for a particular trade size category: "1" if 100 shares ≤ trade size ≤ 499 shares; "2" if 500 shares ≤ trade size ≤ 1,999 shares; "3" if 2,000 shares ≤ trade size ≤ 4,999 shares; "4" if 5,000 shares ≤ trade size ≤ 9,999 shares; "5" if 9,999 shares < trade size.

	Coefficient on $algo\_trad_{it}$					Coefficients on Control Variables			
	Q1	Q2	Q3	Q4	Q5	$share\_turnover_{it}$	$volatility_{it}$	$1/price_{it}$	$ln\_mkt\_cap_{it}$
Panel A: Quoted Spread, Quoted Depth, and Effective Spread									
$qspread_{it}$	−0.53**	−0.42**	−0.43	−0.21	9.92	−2.81**	0.90**	108.40**	−3.61**
	(−3.23)	(−2.21)	(−1.44)	(−0.06)	(1.22)	(−2.98)	(9.71)	(7.42)	(−2.28)
$qdepth_{it}$	−3.49**	−1.43	−1.99	15.60	0.61	−5.22	−1.64*	−3.44	12.01
	(−2.51)	(−1.16)	(−1.07)	(0.39)	(0.19)	(−0.64)	(−1.86)	(−0.02)	(0.82)
$espread_{it}$	−0.18**	−0.32**	−0.35	−1.67	4.65	−1.01**	0.69**	72.77**	−1.30
	(−2.67)	(−2.23)	(−1.56)	(−0.42)	(1.16)	(−2.30)	(9.39)	(10.80)	(−1.46)

(continued)

Table III—*Continued*

	Coefficient on $algo\_trad_{it}$					Coefficients on Control Variables			
	Q1	Q2	Q3	Q4	Q5	$share\_turnover_{it}$	$volatility_{it}$	$1/price_{it}$	$ln\_mkt\_cap_{it}$
Panel B: Effective Spread by Trade Size Category									
$espread1_{it}$	−0.12**	−0.14**	−0.17	−1.83	4.99	−0.83**	0.28**	50.45**	−1.10
	(−3.06)	(−2.02)	(−1.09)	(−0.44)	(1.20)	(−2.70)	(3.91)	(12.43)	(−1.61)
$espread2_{it}$	−0.22**	−0.30**	−0.41	−4.21	4.21	−1.62**	0.43**	53.84**	−2.24
	(−3.25)	(−2.61)	(−1.62)	(−0.45)	(1.16)	(−2.68)	(2.80)	(8.15)	(−1.58)
$espread3_{it}$	−0.25**	−0.26	−0.66	−3.27	6.89	−1.85**	0.64**	61.48**	−1.96
	(−2.76)	(−1.42)	(−1.57)	(−0.36)	(1.13)	(−2.67)	(4.03)	(7.55)	(−1.29)
$espread4_{it}$	−0.11	−0.29	−0.24	643.14	−3.27	28.99	−10.35	224.78	93.04
	(−1.31)	(−0.90)	(−0.62)	(0.00)	(−0.38)	(0.00)	(−0.00)	(0.01)	(0.00)
$espread5_{it}$	−0.04	−0.32	−0.36	1.45	10.48	−0.21	1.05**	73.01**	0.12
	(−0.45)	(−1.04)	(−0.86)	(0.26)	(0.25)	(−0.25)	(4.01)	(8.06)	(0.05)
Panel C: Spread Decompositions Based on 5-Min and 30-Min Price Impact									
$rspread_{it}$	0.35**	0.76**	1.03**	14.25	15.88	3.13*	−1.06**	45.81**	5.05
	(3.52)	(3.97)	(2.06)	(0.46)	(1.36)	(1.92)	(−2.15)	(4.14)	(1.18)
$adv\_selection_{it}$	−0.53**	−1.07**	−1.39**	−15.51	−11.21	−4.12**	1.76**	26.65*	−6.30
	(−3.56)	(−4.08)	(−2.06)	(−0.47)	(−1.33)	(−2.23)	(3.28)	(1.84)	(−1.34)
$rspread\_30m_{it}$	0.33**	0.47*	0.91	11.11	12.63	2.69**	−2.33**	52.24**	2.83
	(2.82)	(1.94)	(1.61)	(0.47)	(1.31)	(1.98)	(−5.99)	(4.21)	(0.83)
$adv\_selection\_30m_{it}$	−0.51**	−0.81**	−1.27*	−12.60	−8.28	−3.66**	3.02**	20.21	−4.10
	(−3.43)	(−2.76)	(−1.80)	(−0.47)	(−1.25)	(−2.33)	(6.91)	(1.35)	(−1.05)

#observations: 1,082 ∗ 167 (stock ∗ day)
F test statistic of hypothesis that instruments do not enter first-stage regression: 7.32 (F(5, 179,587)), p-value: 0.0000

Read:

- More AT narrows spreads in large and medium stocks. In quintile 1, the coefficient on algo_trad_{it} is about −0.53 for quoted spreads and −0.18 for effective spreads; in quintile 2, the corresponding coefficients are about −0.42 and −0.32.
- Quoted depth declines for the largest stocks (about −3.49), but the paper argues that the spread improvement dominates the depth loss for economically relevant order sizes.

- The strongest evidence is concentrated in the large-cap end of the market, which is exactly where AT was most prevalent at the time.
- Economically, the effect is large. Using the within-stock standard deviation of the AT variable (**4.54**), the quintile-1 coefficient implies about **2.41 bps** narrower quoted spreads, nearly half of the mean quoted spread.

5.5 Table III Panel B: Which trade sizes benefit?

Read:

- The spread improvement is concentrated in trades below **5,000 shares**.
- For the large-cap quintiles, the coefficients are negative and significant for trades in the **100–499**, **500–1,999**, and **2,000–4,999** share bins.
- The point estimates for larger trades are generally in the same direction but imprecise.
- This is consistent with AT primarily improving the execution environment for the smaller trade slices that algorithms actually submit, rather than for large parent orders observed as a whole.

5.6 Table III Panel C and Figure 3: Why do spreads fall?

Read:

- The key mechanism is lower adverse selection. In quintiles 1 through 3, the adverse-selection component falls sharply: about −0.53, −1.07, and −1.39 at the 5-minute horizon.
- Realized spreads *increase* at the same time, by about **0.35**, **0.76**, and **1.03** bps in quintiles 1 through 3.
- This is the paper's most surprising result. AT improves overall liquidity not because liquidity suppliers earn less, but because they lose much less to informed or better-timed liquidity demanders.
- Figure 3 suggests that the realized-spread increase is temporary: liquidity-provider revenues rise around the rollout, then decline later in 2003, consistent with temporary market power by incumbent algorithms.

5.7 Table IV: Nonspread outcomes

Read:

- Specialist participation falls for the largest stocks. In quintile 1, the coefficient is about −0.59, which suggests that specialists are *not* the main beneficiaries of the rents created by autoquote.
- Trade size declines for large and medium stocks: about −2.04 thousand dollars in quintile 1 and −0.80 in quintile 2.
- This supports the common view that AT encourages order slicing and the execution of large parent orders through many smaller trades.

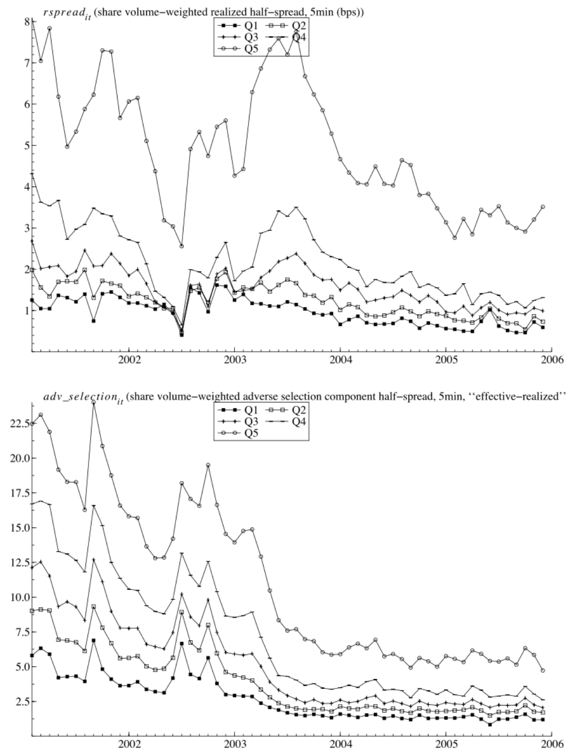

Figure 3. Spread decomposition into liquidity supplier revenues and adverse selection. These graphs depict the two components of the effective spread: (i) realized spread and (ii) the adverse selection component, also known as the (permanent) price impact. Both are based on the quote midpoint 5 minutes after the trade. Results are graphed by market-cap quintile, where Q1 is the largest-cap quintile.

Table IV

Effect of Algorithmic Trading on Nonspread Variables

This table regresses nonspread variables on our algorithmic trading proxy. It is based on daily observations from December 2002 through July 2003, which covers the phase-in of autoquote. The nonsynchronous autoquote introduction instruments for the endogenous $algo\_trad_{it}$ to identify causality from algorithmic trading to these nonspread variables. The specification is

$$M_{it} = \alpha_i + \gamma_t + \beta A_{it} + \varepsilon_{it},$$

where M_{it} is a nonspread variable for stock i on day t, and A_{it} is the algorithmic trading measure. Fixed effects and time dummies are included. There are separate regressions for each size quintile, and t-values in parentheses are based on standard errors that are robust to general cross-section and time-series heteroskedasticity and within-group autocorrelation (see Arellano and Bond (1991)). */** denote significance at the 95%/99% level.

	Coefficient on $algo\_trad_{it}$				
	Q1	Q2	Q3	Q4	Q5
$share\_turnover_{it}$	0.04	−0.07*	0.01	−0.20	−0.36**
	(1.02)	(−1.77)	(0.07)	(−0.26)	(−2.89)
$trades_{it}$	0.58**	−0.01	−0.01	−0.51	−0.15**
	(2.60)	(−0.23)	(−0.15)	(−0.33)	(−2.60)
$trade\_size_{it}$	−2.04**	−0.80**	−0.33	2.27	−0.22
	(−4.64)	(−3.23)	(−0.69)	(0.20)	(−0.60)
$specialist\_particip_{it}$	−0.59**	−0.23	−0.92	−13.24	−1.89**
	(−2.22)	(−1.24)	(−1.43)	(−0.29)	(−2.02)

#observations: 1,082 ∗ 167 (stock ∗ day)
F test statistic of hypothesis that instruments do not enter first-stage regression:
5.88(F(5,179,607)), p-value: 0.0000

- Trades per minute rise for the largest stocks, which fits the idea that the market becomes more active at the micro level even if aggregate turnover does not move much.

5.8 Figure 5 and Table V: Price discovery shifts from trades to quotes

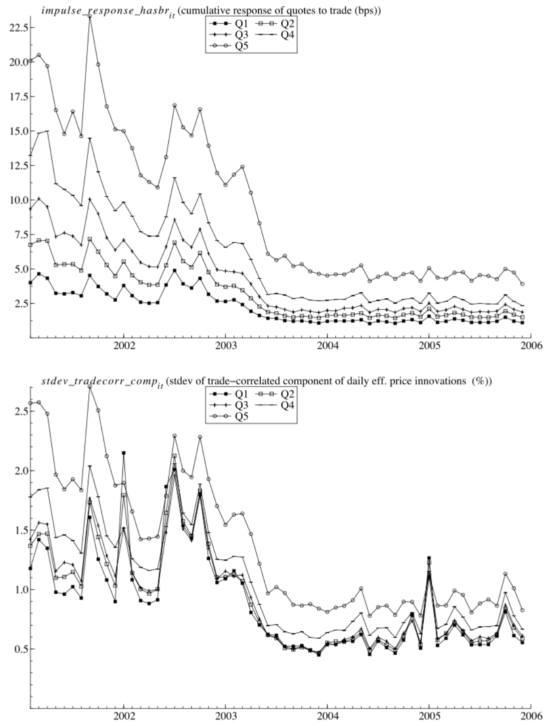

Figure 5. Trade-correlated and trade-uncorrelated information. These graphs illustrate the estimation results of the Hasbrouck (1991a,b) VAR model for midquote returns and signed trades. The top graph illustrates the time-series pattern of the long-term response of the midquote price to a unit impulse in the signed trade variable. The bottom two graphs illustrate the decomposition of the daily percentage variance of changes in the efficient price into a trade-related (*stdev_tradecorr_comp$_{it}$*) and trade-unrelated (*stdev_nontradecorr_comp$_{it}$*) component (see Hasbrouck (1991a,b) for details on the methodology). The graph includes the autoquote sample period, which runs from December 2002 through July 2003. Results are graphed by market-cap quintile, where Q1 is the largest-cap quintile.

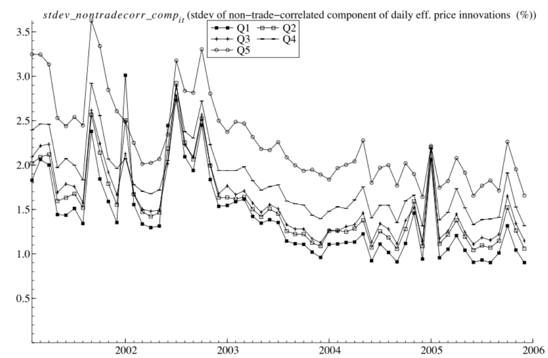

Figure 5. Continued

Read:

- The permanent price impact of trades falls with AT. In Table V Panel A, the coefficient on the Hasbrouck impulse response is about −0.54 in quintile 1, −1.21 in quintile 2, and −1.44 in quintile 3.
- The trade-related component of efficient-price variance also falls. In Panel B, the coefficient is about −0.22 in quintile 1 and −0.26 in quintile 2.
- At the same time, the nontrade-related component rises by about **0.12** in the two largest quintiles.
- The interpretation is central: after AT increases, prices incorporate more information through quote updates and less through actual trades. Quotes become more informative, and trades become less informative.

5.9 Economic magnitude from the price-discovery results

Read:

Table V

Effect of Algorithmic Trading on Permanent Price Impact and Efficient Price Variance Composition

This table regresses the permanent price response to a trade and the two components of efficient price variance on our algorithmic trading proxy. The daily sample extends from December 2002 through July 2003, which covers the phase-in of autoquote. A Hasbrouck VAR model on midquote returns and signed trades is estimated in order to identify the long-term price impact of a trade ($impulse\_response\_hasbr_{it}$) and the trade-related ($stdev\_tradecorr\_comp_{it}$) and trade-unrelated ($stdev\_nontradecorr\_comp_{it}$) components of the daily percentage variance of changes in the efficient price (see Hasbrouck (1991a,b) for details). For the regressions, the nonsynchronous introduction of autoquote is used as an instrument for the endogenous $algo\_trad_{it}$ to identify causality from algorithmic trading to these variables. The specification is

$$M_{it} = \alpha_i + \gamma_t + \beta A_{it} + \delta X_{it} + \varepsilon_{it},$$

where M_{it} is the dependent variable for stock i on day t, A_{it} is the algorithmic trading measure, and X_{it} is a vector of control variables, including share turnover, volatility, 1/price, and log market cap. Fixed effects and time dummies are included. Coefficients for the control variables and time dummies are quintile-specific. Market cap–weighted coefficients are reported for the control variables. Control variables are excluded from the Hasbrouck component regressions. There are separate regressions for each size quintile, and t-values in parentheses are based on standard errors that are robust to general cross-section and time-series heteroskedasticity and within-group autocorrelation (see Arellano and Bond (1991)). */** denote significance at the 95%/99% level.

	Coefficient on $algo\_trad_{it}$					Coefficients on Control Variables			
	Q1	Q2	Q3	Q4	Q5	$share\_turnover_{it}$	$volatility_{it}$	$1/price_{it}$	$ln\_mkt\_cap_{it}$
Panel A: Effect of Algorithmic Trading on the Long-Term Price Impact of a Trade (bps)									
$impulse\_response\_hasbr_{it}$	−0.54**	−1.21**	−1.44**	−18.92	−11.93	−4.90**	1.11*	16.66	−7.83
	(−3.50)	(−4.05)	(−2.08)	(−0.46)	(−1.36)	(−2.20)	(1.66)	(1.15)	(−1.35)
Panel B: Effect of Algorithmic Trading on Trade- and Nontrade-Correlated Component of Daily Efficient Price Variance (%)									
$stdev\_tradecorr\_comp_{it}$	−0.22**	−0.26**	−0.30*	−3.40	−0.57**				
	(−2.62)	(−3.08)	(−1.69)	(−0.29)	(−2.73)				
$stdev\_nontradecorr\_comp_{it}$	0.12**	0.13**	0.13	1.04	0.13				
	(2.48)	(2.36)	(1.47)	(0.28)	(1.12)				

#observations: 1,082 ∗ 167 (stock ∗ day)

F test statistic of hypothesis that instruments do not enter first-stage regression: Panel A: 7.32(F(5, 179, 587)), p-value: 0.0000; Panel B: 5.88(F(5, 179, 607)), p-value: 0.0000

- The within-stock standard deviation of the AT measure is **4.54** during the rollout sample.
- Multiplying this by the quintile-1 coefficient on the trade-correlated standard deviation (-0.22) implies about a **1 percentage point** decline in the daily standard deviation of trade-related price discovery for large-cap stocks.
- The paper emphasizes that this is of the same order of magnitude as the actual level of trade-correlated price discovery in the data.
- So the change is not just statistically significant. It is economically large.

6 Short summary

- **Core finding.** Algorithmic trading improves liquidity on the NYSE, especially for large-cap stocks.
- **Mechanism.** AT helps liquidity suppliers update quotes faster and avoid being picked off, which lowers adverse selection and shifts price discovery away from trades and toward quotes.
- **Identification.** The causal evidence comes from the staggered 2003 introduction of autoquote, used as an instrument for the intensity of algorithmic trading.
- **Important nuance.** AT does not initially improve liquidity through lower liquidity-provider revenues. Realized spreads actually rise for a while, suggesting temporary rents or market power for early algorithms.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that a rise in AT caused by a market-design innovation can improve liquidity.
- The most robust improvements are narrower quoted and effective spreads, lower permanent price impact of trades, and a lower trade-related component of price discovery.

- The evidence therefore supports the view that, at least in normal market conditions, AT can enhance quote informativeness and reduce trading costs.

7.2 Economic intuition

- Limit orders give other traders an option to trade. The more quickly a liquidity supplier can update or cancel those orders when information changes, the less costly that option becomes.
- AT lowers the marginal cost of monitoring order flow and public information. Algorithms can revise quotes almost immediately when market conditions change.
- As a result, prices adjust more through quote updates and less through trade executions that "pick off" stale quotes. That is why adverse selection falls.

7.3 Caveats

- The paper studies a relatively calm period. It does not tell us whether algorithmic liquidity is equally resilient during severe market stress or sharp market declines.
- The data do not identify which specific traders are algorithmic, so the paper cannot directly map the gains from autoquote to a precise class of firms.
- The exclusion restriction—that autoquote affects liquidity only through AT—is economically plausible but ultimately not directly testable.

7.4 Takeaway

This paper is best understood as an early causal microstructure study of trading technology. Its main message is that algorithmic trading did not simply coincide with better liquidity on the NYSE; it *helped cause it*. The improvement comes less from tougher price competition among liquidity suppliers than from a reduction in stale quotes and adverse-selection losses. In that sense, the paper's deepest contribution is not only that AT narrows spreads, but that it changes *how* information gets into prices: less through trades, more through quotes.